An Exact Order-10 Certificate for the Erdős Triangle-Free Bipartization Conjecture

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Abstract

For a graph G, let $\mathrm{bip}(G) = |E(G)| - \mathrm{MaxCut}(G)$ be the minimum number of edges whose deletion makes G bipartite. We give an exact, independently replayable order-10 flag-algebra certificate which, combined with the per-root-MaxCut envelope and blow-up transfer of Ferudun and the density-tail theorem of Balogh, Clemen and Lidický, proves

$$\mathrm{bip}(G) \leq |V(G)|^2/25$$

for every finite triangle-free graph G. The new exact dual has objective $\delta_\star < -9/10000 < 0$. It consists of 10,188 finitely described K7, K8 and rooted-Horn rows, together with the public manifestly positive-semidefinite Gram functional. A verifier rebuilds all 11,560,170 nonzero matrix entries and checks every dual residual in exact rational arithmetic. This manuscript and certificate are a draft for external review.

1 Introduction

For a finite simple graph G, write

$$\mathrm{bip}(G) = \min\{|F| : F \subseteq E(G),\ G - F \text{ is bipartite}\} = |E(G)| - \mathrm{MaxCut}(G).$$

Erdős, Győri and Simonovits asked whether every triangle-free graph G on N vertices satisfies

$$\mathrm{bip}(G) \leq \frac{N^2}{25}. \tag{1}$$

The constant is sharp: for $N = 5n$, the balanced blow-up $C_5[n]$ has $\mathrm{bip}(C_5[n]) = n^2 = N^2/25$.

Balogh, Clemen and Lidický proved the exact constant in low- and high-edge-density tails, and a slightly weaker general bound [1]. Ferudun developed an order-10 per-root-MaxCut/Horn/Gram certificate for the remaining middle-density band and, using integrality, proved the conjectured equality for balanced C_5 multiples through $N = 200$ [4]. A later exact certificate extended this finite range through $N = 240$. The present note completes the same certificate route: an enlarged but still finite row pool has a strictly negative exact dual objective. Consequently no integrality margin is needed.

Theorem 1. *Every finite triangle-free graph G on N vertices satisfies*

$$\mathrm{bip}(G) \leq \frac{N^2}{25}.$$

If $5 \mid N$, equality is attained by the balanced blow-up of C_5.

2 Blow-up transfer and the density split

For an integer $t \geq 1$, let $G[t]$ be the graph obtained by replacing each vertex of G by an independent set of size t and every edge by the corresponding complete bipartite graph.

Lemma 2 (Ferudun). *For every finite graph G and every $t \geq 1$,*

$$\mathrm{bip}(G[t]) = t^2 \, \mathrm{bip}(G).$$

Proof. The cut value of $G[t]$, divided by t^2, is a multilinear function of the fractions $x_v \in [0, 1]$ of each twin class placed on one side. A multilinear function on a cube attains its maximum at a vertex. Thus an optimal cut is constant on every twin class, so $\mathrm{MaxCut}(G[t]) = t^2 \, \mathrm{MaxCut}(G)$. Since $|E(G[t])| = t^2 |E(G)|$, the assertion follows. □

Define

$$d_{\mathrm{mono}}(G) = \frac{2 \, \mathrm{bip}(G)}{N^2}, \qquad d_{\mathrm{edge}}(G) = \frac{2|E(G)|}{N^2}.$$

Lemma 2 makes d_{mono} blow-up invariant. If W_G is the step graphon of G, then

$$d_{\mathrm{mono}}(W_G) = \frac{2 \, \mathrm{bip}(G)}{N^2} \tag{2}$$

exactly; there is no finite-size error.

We use the closed middle band

$$\mathrm{LO} = \frac{1243}{5000} = 0.2486, \qquad \mathrm{HI} = \frac{3197}{10000} = 0.3197.$$

Balogh, Clemen and Lidický prove, for all sufficiently large orders, the bound (1) when the edge density lies below LO or above HI [1, Theorem 1.3]. Applying their theorem to $G[t]$ and then using Lemma 2 transfers both exact-constant tail bounds to every finite G whose graphon-normalized density is strictly outside the closed middle band. The two boundary values will be included in the certificate band.

3 The finite relaxation

We use the order-10 relaxation of [4]. Its probability vector $q = (q_s)$ is indexed by the 12,172 triangle-free order-10 states and satisfies $q_s \geq 0$, $\sum_s q_s = 1$, and $\mathrm{LO} \leq \langle e, q \rangle \leq \mathrm{HI}$. A fixed manifestly positive-semidefinite moment functional gives $\langle m, q \rangle \geq 0$.

For each of the 107 triangle-free K7 root types and each Boolean profile rule c, the rule defines a genuine global two-coloring and hence a valid per-root MaxCut row. The analogous K8 construction has 410 root types. Writing u^7_σ, u^8_τ for the root envelope variables, the finite LP contains selected valid rows

$$u^7_\sigma \leq \langle g_{\sigma,c}, q \rangle, \qquad u^8_\tau \leq \langle h_{\tau,c}, q \rangle,$$

together with

$$\eta \leq \sum_\sigma u^7_\sigma, \qquad \eta \leq -\frac{2}{25} + \sum_\tau u^8_\tau.$$

It also contains rooted-Horn inequalities. Their validity is the copositivity inequality

$$\left(\sum_{i=0}^{4} x_i \right)^2 - 4 \sum_{i=0}^{4} x_i x_{i+1} \geq 0 \quad (x_i \geq 0),$$

with indices modulo 5; equivalently it is the Motzkin-Straus inequality for C_5. Every genuine band step graphon induces a feasible point. Therefore the primal optimum upper-bounds $d_{\mathrm{mono}}(W) - 2/25$ on the band. A feasible dual of objective δ proves

$$d_{\mathrm{mono}}(W) \le \frac{2}{25} + \delta. \tag{3}$$

Only dual feasibility, not numerical optimality, is needed.

4 The exact negative certificate

The new certificate uses the row counts in Table 1.

Row family	Number of rows
K7 per-root MaxCut	2,790
K8 per-root MaxCut	2,385
Rooted Horn	5,013
Total	10,188

Table 1: Exact descriptor pool.

All non-mass multipliers are nonnegative integers over the common denominator $D = 10^{12}$. The static numerators, in the order high-density, low-density, fixed Gram, K7 leg and K8 leg, are

$$(0, 0, 53956267978680, 1, 999999999999).$$

In particular, the envelope legs sum to D exactly. For every K7 root the sum of its rule multipliers is at least $1/D$, and for every K8 root it is at least $(D-1)/D$. No root repair was needed after rationalization.

Let ρ be the exact maximum of the 12,172 rational coordinate loads. The resulting exact dual objective is stored as a numerator-denominator pair in the certificate. Direct integer comparison gives

$$\delta_\star = -9.878886951679021\ldots \times 10^{-4} < -\frac{9}{10000} < 0. \tag{4}$$

The minimum q-coordinate residual is exactly zero, attained at state 46. Thus the certificate is exactly feasible and (3) gives

$$d_{\mathrm{mono}}(W) < \frac{2}{25} \tag{5}$$

for every triangle-free step graphon in the closed middle band.

5 Independent exact replay

The primary certificate is a transparent JSON file. The independent verifier does not import the LP generator and does not trust a stored floating matrix. It performs the following checks.

1. It validates all descriptor syntax and reconstructs every K7, K8 and Horn functional from exact finite combinatorial data.

2. It builds an integer matrix of shape $10{,}188 \times 12{,}172$ with 11,560,170 nonzero entries.

3. It checks nonnegativity, the exact envelope-leg sum, and all 517 per-root multiplier lower bounds.

4. It computes the sparse transpose product by base-100,000 digit splitting, with an a priori signed-64-bit bound, and audits nine columns by direct Python-integer dot products.

5. It reconstructs all coordinate loads and the objective as exact rational numbers and verifies that the minimum residual is zero and that $\delta_\star < 0$.

Separately, the fixed moment vector is reconstructed from 87 nonnegative rational rank-one Gram atoms and matches all 12,172 coordinates exactly. Both the JSON and pickle certificate representations pass the replay. The terminal output is

```
EXACT_DUAL_REPLAY_OK
descriptors=10188 nnz=11560170 digits=3
digit_product_audit=9/9
min_q_residual=0 at state=46
objective ~= -9.878886951679021e-04
global_closure=True
GLOBAL_CONCLUSION: beta(G)<=N^2/25 for every finite triangle-free graph
```

6 Proof of Theorem 1

Let G be a triangle-free graph on N vertices.

If $d_{\text{edge}}(G) < \text{LO}$ or $d_{\text{edge}}(G) > \text{HI}$, the transferred Balogh-Clemen-Lidický tail theorem gives $\text{bip}(G) \le N^2/25$.

Otherwise $\text{LO} \le d_{\text{edge}}(G) \le \text{HI}$. Applying the exact band certificate to W_G and using (2) and (4),

$$\frac{2\,\text{bip}(G)}{N^2} = d_{\text{mono}}(W_G) \le \frac{2}{25} + \delta_\star < \frac{2}{25}.$$

Hence $\text{bip}(G) < N^2/25$, which is stronger than the required inequality in the middle band. These cases cover every possible density, including the two band boundaries. Finally, $C_5[n]$ shows equality for $N = 5n$. □

7 Reproducibility and status

The ancillary package contains the exact JSON and pickle certificates, the independent replay, the moment-vector verifier, the floating source dual, the descriptor pool, and scripts that regenerate the large K7 exact-count cache. The compact certificate archive omits that cache. A separate full replay bundle includes the cache and all four public flag-data files needed by the verifiers; it is about 20 MB compressed and about 1.8 GB unpacked. The public repository runs the full replay in continuous integration.

The finite certificate and fixed Gram vector have been independently replayed inside the present computational environment in exact arithmetic. At the time of writing, the argument has not received external peer review. It should therefore be treated as a reproducible computer-assisted proof draft pending external verification.

References

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[4] A. Ferudun, *The Erdős $n^2/25$ max-cut conjecture for small multiples of five, via a per-root-MaxCut envelope and blow-up integrality*, arXiv:2606.28041v1 (2026), `https://arxiv.org/abs/2606.28041`.

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